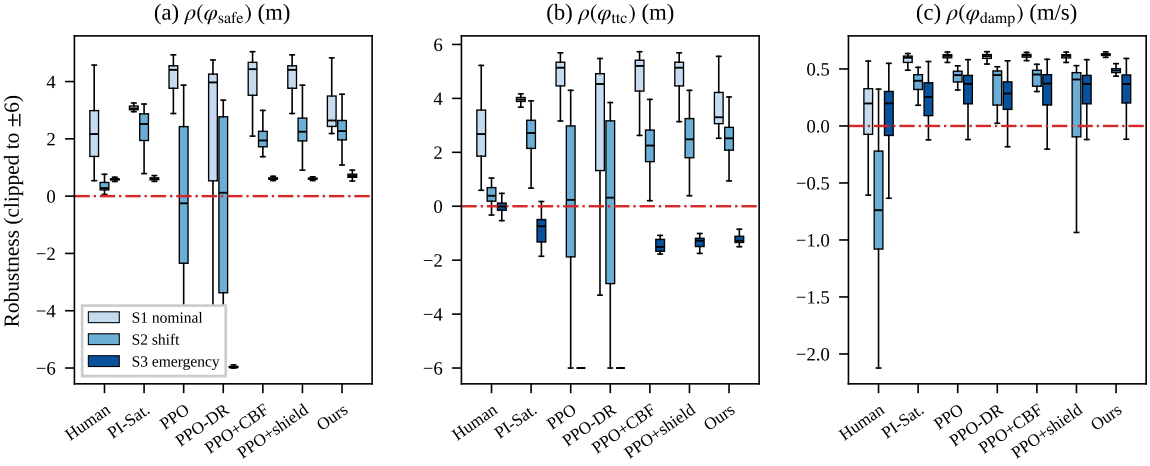
(a) $\rho(\varphi_{\text{safe}})$ (m)
(b) $\rho(\varphi_{\text{ttc}})$ (m)
(c) $\rho(\varphi_{\text{damp}})$ (m/s)
Robustness (clipped to ±6)
S1 nominal
S2 shift
S3 emergency
Human
PI-Sat.
PPO
PPO-DR
PPO+CBF
PPO+shield
Ours